PRODUCT SPECIFICATION · VERSION 2.0

Kang Automation
Master Product Roadmap

A modular, approval-driven creative automation platform for video editing, motion graphics and creative design — refined edition.

DOCUMENT	VERSION	DATE	STATUS	SUPERSEDES
Master Roadmap	**2.0 (Refined)**	**29 Oct 2026**	**Draft for Review**	**v1.0**

WHAT CHANGED IN V2.0

1. Structure. 17 overlapping sections consolidated into 17 blocks with a single canonical source for each rule (lifecycle, QA, Definition of Done no longer repeat each other).

2. Traceability. Every lifecycle step now has a stage ID (S0–S12); every feature states its tier and target milestone; a workflow → feature → milestone → gate matrix was added.

3. Measurability. A KPI set with numeric targets and a measurement method was added; vague acceptance language was replaced by testable statements.

4. Planning. Milestones M0–M7 now carry dependencies, indicative duration, critical path, Definition of Ready and a named owner role.

5. Corrections. Typo `AF04` → `F04`; missing `approvals.json` and `job_queue.json` artifacts added; face-detection threshold given a real sampling specification; dead "Reserved ID" row converted to an explicit reservation note.

6. New sections. Risk register, open decisions, glossary, change-control rule, feature status snapshot.

This document specifies intended behaviour and acceptance evidence. It is **not** a certificate of completion: no capability is considered delivered until implementation evidence exists and is recorded in the QA report.

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Reading guide

Sections 1–2 answer *what are we building and for whom*. If a proposal does not serve a numbered workflow or a KPI, it is out of scope by default.

Sections 3–7 are the build contracts: stages, gates, artifact schemas, router rules, transcript rules and features. These are the documents an implementer works from.

Sections 8–11 define what "done" means and on which hardware it may be claimed.

Sections 12–17 are the plan, the risks, the checklist and the reference appendices.

> **HONESTY RULE (APPLIES TO EVERY SECTION)**
>
> Every status claim must resolve to one of five values: `PASS · FAIL · WARN · NOT_TESTED · NOT_APPLICABLE`. A capability that has never been executed on its target hardware or application is **NOT_TESTED**, never "working". At the time of writing, **no feature in this roadmap carries implementation evidence**; the whole catalogue is specification-stage.

1 Product definition, principles and success metrics

Kang Automation turns voiceovers, camera footage, references, images and creative briefs into reviewed creative assets and portable editing packages — inside validated, approval-gated workflows. Final creative control stays with the user, who assembles in Premiere.

1.1 Primary outcome — the single acceptance target

The product is judged first on one path (**Workflow G**):

```
10–15 min Hindi/Hinglish voiceover → validated input → transcript + coverage review → accepted transcript → scene & edit plan → Gate 1 → approved visuals + captions → preview → Gate 2 → portable Premiere package → QA evidence
```

Success means: the user opens the package in Premiere, finds **no missing media and no missing fonts**, can still edit the elements declared editable, and can trace every decision back to a logged approval. Everything else in this roadmap is either an input to this path or an extension built after it works.

1.2 Principles

PRINCIPLE	WHAT IT MEANS IN PRACTICE
One core, many executors	A single workflow core drives FFmpeg, the motion renderer, design tools and NLE adapters. Executors may be swapped; the contract may not.
Modular with declared contracts	Each module declares inputs, outputs and tests. A module that cannot state its output schema is not a module.
Human control at judgement points	Subjective edits and all external or paid actions require explicit approval. Automation proposes; the reviewer disposes.
Evidence over claims	Acceptance requires artifacts, logs, versions, tests and QA records. Process exit code 0 is not evidence.
Immutability and isolation	Source files are read-only and content-hashed; every run gets a unique ID and an isolated workspace.
Portability	Outputs carry explicit asset manifests, rights and editability claims, and survive being moved, zipped and relinked.
No silent substitution	Fallbacks (safe crop, placeholder, approved substitute) are visible in the UI, the manifest and the QA report.
Publishing is out of band	Nothing is published, posted or uploaded automatically. Publishing is a separate, explicitly approved action.

1.3 Supported input families

FAMILY	EXAMPLES	SHARED PROCESSING	VISUAL STRATEGY
Visual strategy (audio-only)	MP3 / WAV voiceover	Probe, transcript, correction, captions, packaging	Approved scenes and assets, brand card, optional motion
Face video	Talking head, presenter	Shared processing + original footage	Preserve footage; optional subject-follow reframe (F06) for 9:16 and 1:1
No-face video	Screen recording, B-roll, slides	Shared processing + original footage	Safe crop or pad; F06 off by default
Mixed / uncertain	Face appears in some segments only	Shared processing + segment flags and user review	Per-segment branch decision, explicit override required
Image / design	Screenshot, logo, poster brief	Validation, brand and rights, manifest	Design, style extraction, optional motion

1.4 Non-goals

- **No guaranteed virality and no exact replication** of a reference edit or a competitor's style.
- **No editable layers from flattened MP4** — flattened video is a deliverable, not a source.
- **No face identity recognition, face swap or voice cloning** commitments, at any milestone.
- **No URL-only sourcing**: references must be authorized files, not links.
- **No "free compute for everything"**: not all tools, models, fonts or media are free or usable commercially.
- **No AI visual presented as fact**: generated visuals are illustrations, never evidence.

1.5 Success metrics

Targets below apply to Workflow G unless stated. Baselines are recorded on the first natural 10–15 min source; targets become commitments from M3 onward.

METRIC	TARGET	DEFINITION AND MEASUREMENT
Time to first approved preview	≤ 60 min human effort	Sum of reviewer interaction time from intake to Gate 2. Machine render wait excluded and reported separately.
Manual interventions per run	≤ 10, tracked	Count of hand-edits to plans, cuts, captions or assets after automation. Logged in `run_manifest.json`.
Transcript accuracy after correction	≥ 98% word-level	100-word Quick Sample; ≥ 90% before correction. Computed as `1 − WER`.
Unresolved voiced gaps > 2 s	0	`coverage_report.json` must show every such gap resolved or explicitly reviewed.
Caption sync error	≤ ±100 ms	Against human-marked reference timings, minimum 20 markers per run.
Broken Devanagari glyphs	0	Full visual audit of a 3-minute captioned segment across all three ratios.
Package import success	100% (n = 5)	Five relocations (moved, zipped/extracted, drive-letter changed) opened in the declared Premiere build.
Re-run rate after Gate 2	≤ 20%	Percentage of runs requiring re-execution because preview approval failed.
Paid spend per run	0 by default, capped when enabled	Paid calls are off by default; when enabled, a per-run cap and a cost record are mandatory.
Checkpoint recovery	100%	Every cancel/failure test must resume from the last validated checkpoint with no partial artifact accepted as cache.

> **WHY THESE AND NOT "QUALITY"**
>
> "Good edit" cannot be tested. Each metric above maps to an artifact that exists after every run, so a reviewer can verify the number without asking the builder anything. Any new KPI must name its artifact in the same sentence.

2 Scope: workflows A–I and the primary milestone path

Nine workflows share one core. They differ in routing, optional features and the gate they exit at — not in their foundations. Each workflow now names the milestone where it must first pass.

ID	WORKFLOW	PIPELINE	GATE / OUTPUT	FIRST PASS
A	Talking head → Reel	Router → transcript / coverage / correction → approved cuts → F04 → F06 → F01/F02 → preview → F05 export	Cut + preview approval; 9:16 output, captions, manifest	M4
B	Screenshot → Motion	Validate image → F19 → style approval → planner → F20 template → render / QA	Approximate style label; motion output	M5

ID	WORKFLOW	PIPELINE	GATE / OUTPUT	FIRST PASS
C	Podcast → Shorts	Long video → transcript → F07 candidates → selection → optional F03/F10/F06/F01 → batch QA	Candidate + asset rights approval	M6
D	Brief → Design pack	Brief + assets → F16 → F13 and/or F15 → optional F14 → QA / package	Variant selection; social and print outputs	M5
E	Lyrics → Kinetic text	Correct lyrics → Hindi-verified alignment → F21 / F20 → optional approved F23 → preview / QA	Alignment + preview approval	M6
F	Reference-inspired edit	Authorized reference → F09 → own content + planner → Gate 1 → selected F03/F10/F01/F20 → preview	Estimates only; no protected-content copying	M6
G	**Voiceover → Premiere**	Router → ASR / coverage → correction → accepted transcript → scenes → Gate 1 → visuals + captions → preview → package → QA → import	Portable package + actual Premiere import evidence	**M3**
H	Image operation	Validate image → operation (e.g. F14) → QA → export	Transcript and EDL stages marked NOT_APPLICABLE	M5
I	Batch	Job list → workflow → QA / status → batch report	Failure isolation; resumable checkpoints	M7

"First pass" = the milestone at which the workflow must complete end-to-end with evidence. It is a gate, not a release date.

2.1 Workflow G in detail (the path that must work before anything else)

STEP	ACTION	OUTPUT	EXIT CONDITION
G1	Intake media, brief, output ratio	Run ID, hashed source copy, isolated workspace	Probe succeeds; source hash recorded
G2	Route (audio-only expected)	`input_profile.json`, router decision + reason	Decision accepted or overridden by user
G3	ASR draft + coverage analysis	Raw transcript, word timings, raw SRT, `coverage_report.json`	Every voiced gap > 2 s listed
G4	Correction review	Accepted transcript (canonical), diff history	Reviewer signs off; accuracy sample ≥ 98%
G5	Plan scenes, cuts, assets, captions	`scenes.json`, `cuts.json`, `assets.json`, `master_edl.json`	Schema valid; no unresolved no-match
G6	**Gate 1** — approve plan	`approvals.json` bound to plan hash	Explicit approval recorded
G7	Execute: visuals, captions, assembly	Rendered segments, SRT/ASS, preview video	Atomic outputs; no partial artifacts
G8	Preview QA	`qa_report.json` (sync, glyphs, loudness, safe zones, rights)	No P0/P1 defect open
G9	**Gate 2** — approve preview	Approval bound to settings hash	Preview approved as DRAFT → FINAL candidate
G10	Final render + package + full QA	MP4, SRT/ASS, XML/package, manifest, logs	Full decode; duration ±1 frame
G11	Premiere import test	Import log, missing-media report	Opens in declared build with zero missing media/fonts
G12	Archive / checkpoint	Archived package, provenance, cost record	Reproducible from inputs + settings

TWO RULES THAT PROTECT THE NARRATION

Narration is preserved by default. Every cut is a suggestion until approved; nothing is removed from the audio without a recorded decision.

Drafts are labelled DRAFT. A preview is never named or presented as FINAL before Gate 2 passes, and failed QA can never be re-labelled as success.

3 Architecture, canonical lifecycle and gates

3.1 Shared core

Input and path validation, timebase conversion, manifests, logging, checkpoints, recovery and QA status are implemented once and reused by every module. A module may not re-implement them.

3.2 Layers

LAYER	RESPONSIBILITY	OUTPUT / RULE
Intake / router	Probe streams, type, dimensions, duration; decide visual branch	`input_profile.json`, router report with reason
Analysis	ASR, speech coverage, scene boundaries, OCR, loudness	Versioned analysis artifacts
Correction	Raw vs corrected text, review flags, accepted revision	Accepted transcript becomes canonical
Planner	Scene plan, Master EDL / design spec, assets, rights	Validated plan + bound asset set
Approval gates	Plan, cuts, assets, paid actions, preview	Approval record bound to a hash
Executors	FFmpeg, motion renderer, design tools, NLE adapters	Atomic outputs only
QA / export	Decode, probe, sync, glyphs, loudness, safe zones, imports	QA report + manifest + deliverables
Recovery / archive	Versions, logs, checkpoints, provenance	Resume from validated checkpoint only

3.3 Canonical lifecycle (single source of truth)

Every document, artifact and QA check in this roadmap references these stage IDs. If a step is not one of S0–S12, it is a module detail, not a stage.

ID	STAGE	PURPOSE	PRODUCES
S0	Intake	Accept media, brief, ratio; freeze inputs	Run ID, hashes, workspace
S1	Validate	Probe, integrity, rights precheck, capability check	Pass/fail with reason
S2	Route	Choose visual branch and confidence	`input_profile.json`
S3	Analyze	ASR, coverage, scenes, audio, face, OCR	Versioned analysis set
S4	Correct / Accept	Human correction; accepted transcript	Canonical transcript + diff log
S5	Plan	Scenes, cuts, assets, captions, EDL / design spec	Validated plan
S6	Gate 1	Approve plan, cuts, assets, rights, paid actions	`approvals.json`
S7	Execute	Render visuals, captions, assembly	Atomic artifacts
S8	Preview QA	Checks on the preview build	QA report (preview scope)
S9	Gate 2	Approve preview and final settings	Approval bound to settings hash
S10	Final render / package	Final outputs and portable package	Deliverables + manifest

ID	STAGE	PURPOSE	PRODUCES
S11	Full QA	Full-scope checks and import test	Final QA report
S12	Export / Archive	Hand-off, provenance, cost, retention	Archive + checkpoint

3.4 Gate semantics

GATE	APPROVES	INVALIDATION RULE
Gate 1 (S6)	Plan, cuts, asset selection, rights posture, any paid or external action	Any change to plan, cuts or asset set invalidates approval; re-approval required before S7
Gate 2 (S9)	Preview quality and final render settings	Any change to settings, captions or assets invalidates approval
Action gate	Paid API calls, uploads, publishing, irreversible deletions	Single-use; scoped to one action and one cost ceiling

> **HASH BINDING**
>
> An approval is not a boolean in a database — it is a signature over the exact artifact set it approved. Store `plan_hash`, `settings_hash`, approver identity, timestamp and the artifact version list inside `approvals.json`. If a downstream artifact's hash no longer matches, the run reverts to `READY_FOR_REVIEW` automatically.

3.5 Status state machine

```
NOT_STARTED → IN_PROGRESS → READY_FOR_REVIEW → APPROVED → EXECUTING
            → QA_PASSED | QA_FAILED → PACKAGED → ACCEPTED

Side states (terminal until reset): BLOCKED · FAILED · CANCELLED · SUPERSEDED
Per-check values:                    PASS · FAIL · WARN · NOT_TESTED · NOT_APPLICABLE
```

Transitions are logged with actor, timestamp and reason. A stage may never jump from `IN_PROGRESS` to `ACCEPTED`; `ACCEPTED` requires a QA report whose applicable checks are all PASS or explicitly accepted WARN.

4 Input router: audio, face, no-face, mixed, image

The router decides the visual branch from stream metadata and face evidence. It never replaces transcription and never guesses silently: every decision carries a reason and a confidence.

4.1 Detection specification

- **Sampling:** decode at 1 frame per second; require at least 10 sampled frames before producing a face decision. Clips shorter than 10 s are routed with confidence capped at 0.6 and flagged for review.
- **Detector:** a tested face detector (MediaPipe Face Detection, or equivalent) with a per-frame confidence floor of 0.5.
- **Decision:** face share = detected frames ÷ sampled frames. ≥ 30% → `video_face`; < 30% → `video_no_face`; 15–30% or intermittent appearance → `mixed` (review required).
- **Probe failure:** route to `unknown`, stop before any processing, preserve the source untouched.
- **Tuning:** the 30% / 15% thresholds are initial values and must be re-tuned against real footage from the target user, with the tuning run logged.

4.2 Routing rules

TYPE	RULE	VISUAL BEHAVIOUR	FALLBACK
`audio_only`	No video stream present	Approved scenes and assets, brand card	Optional motion fill

TYPE	RULE	VISUAL BEHAVIOUR	FALLBACK
video_face	Face in ≥ 30% of sampled frames	Original footage; F06 available for 9:16 and 1:1	Safe crop if tracking is lost
video_no_face	Video present, below threshold	Original footage with safe crop or pad	F06 off by default
mixed	Face in some segments only	Per-segment flags, user review	Explicit override required
unknown	Probe failure or unreadable stream	Stop before processing	Error report; source preserved

4.3 Output ratios

RATIO	AUDIO-ONLY	FACE VIDEO	NO-FACE VIDEO
16:9	Re-layout scenes	Native footage, safe framing	Native footage, safe framing
9:16	Re-layout	Subject-follow crop with documented fallback	Safe crop or pad, approval required
1:1	Re-layout	Subject-follow square crop	Centre crop or pad, approval required

One source may produce several ratios in one run (F05); each ratio is a separate output with its own QA record, not a scale-up of the first.

4.4 Required test fixtures

FIXTURE	PURPOSE	PASS CONDITION
MP3 / WAV voiceover, 10–15 min	Primary path, no video stream	Routes to `audio_only`, zero face calls made
Face clip, mixed framing	Face branch + tracking	Routes to `video_face`; tracking loss produces fallback, not crash
No-face clip (screen recording)	No-face branch	Routes to `video_no_face`; F06 stays off
Mixed clip, face intermittent	Ambiguity handling	Routes to `mixed`; review is requested, never auto-decided
Corrupt / zero-byte file	Safe failure	Fails at S1 with error report; source unchanged
Synthetic fixtures of each class	Deterministic regression tests	Same input always produces same decision + confidence

5 Artifact contracts, naming and run workspace

Artifacts are the product. If two modules disagree about a schema, the schema wins and the module changes. All schemas carry a `schema_version` field; breaking changes require a major bump and a migration note.

ARTIFACT	REQUIRED CONTENT	VALIDATION RULE
run_manifest.json	Run ID, input hashes, schema/config/tool versions, timings, costs, intervention count	Present on every run; contains no secrets
input_profile.json	Streams, duration, dimensions, fps, router decision, confidence, evidence	Decision must carry a reason and the sampling statistics
transcript.json words.json	Raw and accepted transcript, word timings, optional confidence	Monotonic and in-bounds; no invented gap-fill
coverage_report.json	Speech regions, retries, exceptions, reviewer notes	Every suspected voiced gap > 2 s resolved or reviewed
cuts.json	KEEP/REMOVE flags, source range, reason, approval reference	Non-destructive; no removal without an approval record

ARTIFACT	REQUIRED CONTENT	VALIDATION RULE
scenes.json	Scene ID, revisions, source/output times, assets, on-screen text	Valid IDs and times; no unexplained gaps
assets.json **broll.json**	Asset ID, path, hash, type, source, licence, confidence, review state	Portable paths; no-match stated explicitly
brand.json **style.json**	Colours, fonts, logos, margins, safe zones	Font availability and safe-area compliance checked before render
master_edl.json	Source→output mapping, tracks, crop keyframes, asset references	Reject unknown IDs or unsupported effects rather than degrade silently
approvals.json	Gate ID, approver, timestamp, plan/settings hash, artifact versions, scope	Required before S7 and S10; hash mismatch voids approval
qa_report.json	Check ID, stage ID, measured value, status, evidence path	Status limited to PASS / FAIL / WARN / NOT_TESTED / NOT_APPLICABLE
job_queue.json	Job list, per-job state, checkpoint pointer, retry count	Failure isolation: one job's failure never mutates another's state
package_manifest	Relative paths, hashes, editability claims, licences, declared app version	Move / zip / extract / relink test must pass

5.1 Naming and versioning

- Files: `<artifact>.v<N>.json`; N increments, files are never overwritten in place. The manifest lists every version and the one marked canonical.
- Every artifact records the producing module, its version and the config hash that produced it.
- Derived artifacts name their parent version, so the graph can be walked backwards from any deliverable to the source.

5.2 Timebase rules

- Source time and output time are separate fields — never a single overloaded "time".
- Frame rate and timebase are explicit on every timeline artifact; conversion happens only through shared helpers.
- Approved cuts remap captions, scenes, crops and asset placements downstream. The source timeline is never mutated to make an edit look simple.
- Rounding is applied once, at the output boundary, and the rounding rule is recorded.

5.3 Run workspace

```
runs/<run_id>/
  input/         read-only, hashed copies of all sources
  router/        input_profile.json, sampling evidence
  analysis/      scenes, audio, face, OCR outputs
  transcript/    raw, corrected, words, coverage, diffs
  scenes/        scene plans and revisions
  edit/          cuts.json, master_edl.json, approval references
  approvals/     approvals.json + signed review captures
  subtitles/     SRT / ASS per language and ratio
  visuals/       approved assets, brand assets, placeholders
  render/        preview and final renders
  export/        package, manifest, deliverables
  logs/          stage logs, tool versions, timings
  qa/            qa_report.json, evidence screenshots, import logs
```

Folders that do not apply are created empty with a `NOT_APPLICABLE` marker, so the absence of a stage is visible rather than ambiguous.

6 Transcript intelligence and language QA

ASR produces a *draft*, not truth. Hindi/Hinglish spelling, proper nouns, amounts, negations and code-switching are the four failure classes that matter most, and each needs a human-visible check.

6.1 Correction sequence

1. **Preserve the raw layer.** Raw transcript, word timings and raw SRT are written before any correction and are never edited afterwards.
2. **Cross-check speech against words.** Compare detected speech regions with timed words; flag every voiced gap longer than 2 s.
3. **Recover narrowly.** Re-run ASR only on the flagged region. Never infer or fabricate words to close a gap.
4. **Record every edit.** Store before/after text, reviewer, timestamp and revision number in the diff log.
5. **Realign timings** whenever words are inserted, deleted, split or reordered.
6. **Promote to canonical.** The accepted transcript is the only input to captions, scenes and chapters; the raw history remains available.
7. **Invalidate downstream artifacts** whose parent transcript version has changed, and require re-approval where a gate depends on them.

6.2 Targets and measurement

CHECK	TARGET	MEASUREMENT NOTE
Coverage	Every voiced gap > 2 s resolved or reviewed	Short omissions below 2 s may remain; they are listed, not hidden
Quick sample	≥ 90% before correction, ≥ 98% after (100 words)	Words sampled from three positions: start, middle, end
Long audit	≥ 500 reference words reviewed for names, dates, amounts, negations, code-switching	Full-file accuracy is not guaranteed and is not claimed
Metrics	WER = (subs + dels + ins) ÷ reference words; 100 − WER = word accuracy	Report the reference used and who transcribed it
Caption sync	± 100 ms against human-marked audio	Verifies timing only; it does not verify wording
Lyrical sync	± 33 ms where feasible	Singing requires separate validation (F21)
Hindi typography	Zero broken glyphs in audit	Matras, conjuncts, digits and mixed-script runs are all tested

KNOWN LIMITATIONS TO STATE IN THE UI, NOT JUST THE DOCS

Faster-Whisper is a candidate engine but needs a Hindi-verified aligner before any sync claim. Singing alignment is unproven. Word-level confidence is unreliable for code-switched sentences and must not be shown as if it were accuracy.

7 Feature catalogue F01–F23

Tiers determine build order, not importance. **CORE** features are required for the primary path; **EXTENSION** features are valuable but cannot block M3; **RESERVED** means the ID is held with no implementation commitment.

ID	NAME	TIER	MS.	INPUT → APPROACH	OUTPUT	ACCEPTANCE CRITERIA
F01	Animated subtitles	CORE	M3	Timed words + font/style/canvas → ASS/libass or tested browser renderer	SRT/ASS and/or captioned video	Devanagari glyphs and matras intact; sync ±100 ms; inside safe zones; contrast checked on 3 backgrounds
F02	Subtitle presets	CORE	M3	Font, size, outline, position → versioned preset JSON + renderer	Reusable presets and previews	At least 3 sample lines rendered per preset; font verified as installed, not substituted
F03	Silence / filler cuts	CORE	M3	Audio, optional transcript, prefs → FFmpeg analysis + VAD + rules	Non-destructive `cuts.json` + approved edit	No meaningful word auto-deleted; padding honoured; click and breath artefacts checked
F04	Audio cleanup / loudness	CORE	M3	Audio/video + target profile → FFmpeg, optional denoiser	Cleaned media + pre/post reports	−14 LUFS ±1 integrated, true peak ≤ −1.5 dBTP; listening pass for artefacts; measured before and after encode
F05	Multi-format export	CORE	M3	Media/edit plan + ratio profiles → FFmpeg + deterministic layout profiles	16:9, 9:16, 1:1 outputs	Safe crop or pad per ratio; full decode; duration ±1 frame. Captions optional here — F01/F02 not mandatory
F06	Auto reframe	EXT	M4	Video + ratio + subject policy → MediaPipe + tracking, PySceneDetect optional	Crop track/keyframes + reframed output	Jitter measured and thresholded; subject-loss fallback exercised; no frame with zero subject and silent crop
F07	Shorts finder	EXT	M6	Long video + transcript + clip length → transcript rules, optional LLM	Ranked and selected clips	Every clip is a complete thought; all cuts user-reviewed; ranking explainable per clip
F08	Beat sync	EXT	M6	Music + editable segments → librosa + snap logic	Beat-aligned edit plan	Beat detection reported with precision; snap tolerance stated; no drift over 60 s
F09	Reference style / pacing	EXT	M6	Authorized reference clips → PySceneDetect, OCR, audio analysis	Estimated style and pacing profile	Estimate only; never sold as exact replication; reference rights recorded
F10	Visual / B-roll suggestions	EXT	M6	Scene brief + asset library → tags first, optional multilingual embeddings	Suggested placements + rights manifest	No assumed stock download; no-match returned explicitly with a placeholder
F11	Titles / lower thirds	EXT	M6	Text, timing, brand/template → F20 or supported Adobe route	Overlay or render; editable elements declared	Overflow and safe-zone checks; not a prerequisite for M3
F12	Premiere / AE bridge	EXT	M3 / M6	Approved EDL + assets + target app → version-tested adapters	Portable project/package + editability claim	Actual import in declared build; separate tested adapter per app; generic XML ≠ native project

ID	NAME	TIER	MS.	INPUT → APPROACH	OUTPUT	ACCEPTANCE CRITERIA
F13	Thumbnail	EXT	M5	Topic/title, images, brand → templates/layout, optional cutout	Thumbnail variants at 1280×720 minimum, 1920×1080 target	Readability at 20% scale; crop and contrast checks; text not clipped
F14	Background removal / upscale	EXT	M6	Image + operation → rembg, optional Real-ESRGAN	Alpha PNG / upscaled image	Hair and halo artefacts reviewed; model licence verified for commercial use
F15	Poster / banner sizes	EXT	M5	Brief, text, images, logo, sizes → responsive layout, print tools	Social sizes + configured print PDF	Logo aspect preserved; fit and bleed/profile verified per size
F16	Brand kit	CORE	M5	Logo, colours, fonts, margins → `brand.json` + font checks	Reusable brand profile	Single source of truth; safe zones defined; visible fallback when a font is missing
F17	Batch queue	EXT	M7	Job list + features/config → runner, queue, checkpoints	Per-job output, status, resume	Failure isolation; cancel and resume both tested; queue state persisted across restart
F18	Chapters / metadata	EXT	M6	Accepted transcript + timing → rules, optional LLM	Editable chapter, title and description drafts	Timestamps validated against the final timeline; nothing auto-published
F19	Screenshot style extraction	EXT	M5	Screenshot + own content → K-Means, OCR, layout estimate	Lite palette/style JSON or layout estimate	Approximate only; palette labelled as an estimate; no claim of exact extraction
F20	Motion engine	CORE	M5	Text/images/shapes + templates → HTML/CSS/GSAP or Remotion + FFmpeg	Frame-timed motion video or overlay	Versions pinned; decode and frame-repeat test; deterministic re-render
F21	Lyrical / kinetic text	EXT	M6	Lyrics + audio + style → Hindi-verified alignment + F20	Timed kinetic text video	Speech and singing validated separately; human review mandatory before release
F22	Reserved ID	RESERVED	—	No committed input	No deliverable	ID held for future scope; the roadmap has no dependency on it
F23	AI visual assets	EXT	M6	Approved scene prompt, style, size → tested image backend	Image + model, prompt and provenance record	Paid opt-in only; budget cap, cache, licence review, deterministic fallback

7.1 Feature status snapshot

Per the honesty rule, the starting state of every feature is recorded rather than assumed. Update this table only with linked evidence.

FEATURE GROUP	STATUS	WHAT WOULD CHANGE IT
F01–F05 (core edit path)	`NOT_TESTED`	A run on a natural 10–15 min Hindi source with QA report and screenshots
F06, F12 (tracking, NLE bridge)	`NOT_TESTED`	Device-specific tracking test; actual import log from the declared Premiere build
F16, F20 (brand, motion)	`NOT_TESTED`	Repeatable render plus font and licence audit
F07–F11, F13–F15, F18–F21, F23	`NOT_TESTED`	Per-feature acceptance row above, with artifacts attached
F22	`NOT_APPLICABLE`	Reopened only by an explicit scope decision

8 QA and acceptance matrix

One matrix governs all workflows. A check that does not apply is marked NOT_APPLICABLE with a reason — never silently skipped, and never marked PASS by default.

AREA	ACCEPTANCE RULE	EVIDENCE
Input integrity	Probe valid; source unchanged; corrupt input fails safely at S1	ffprobe output, source hash, router report
Transcript	Coverage resolved or reviewed; raw history preserved; accepted version canonical	Coverage report, transcript versions, diff log
Subtitles	No broken glyphs; sync verified against human reference; inside safe zones	Screenshots, sync log, preset version
Cuts	No unintended truncation; only approved cuts applied	`cuts.json`, approval record, listening notes
Audio	−14 LUFS ±1 integrated; true peak ≤ −1.5 dBTP	Pre/post-encode measurements, listening pass
Reframe	Acceptable framing and jitter; fallback documented and exercised	Crop track, failure-case captures
Assets	All IDs resolvable; rights reviewed; no-match stated explicitly	Asset manifest, provenance, licence records
Export	Full decode; duration ±1 frame; no unexplained gaps	ffprobe, playback test per ratio
Premiere	Declared imports succeed; no missing media or fonts; declared elements still editable	App version and build, import log, relink test
Motion	Frames and fonts valid; repeat-render test passes	Render config, licence review, frame check
Security / recovery	No secrets in artifacts; resume only from validated checkpoints	Logs, recovery tests, artifact inventory
Status progression	Each stage moves only through the defined state machine, with evidence per transition	Log and QA reports

8.1 Severity

SEVERITY	DESCRIPTION	ACTION
P0	Data loss, secret leakage, corruption, or any unsafe external action	Stop the release; fix before any further work
P1	Workflow cannot complete, or output is unusable	Blocks acceptance of that milestone
P2	Quality issue with a documented workaround	Disclose in the QA report and track to closure
P3	Enhancement or cosmetic issue	May be deferred; logged as backlog

8.2 Release gate rules

- **Zero P0** at every release, without exception. **Zero P1** from M3 onward.
- A milestone cannot be declared complete while any applicable check is `NOT_TESTED`; it may be released as "milestone incomplete, scope partial" with that status visible.
- `WARN` is allowed only with a named owner, a workaround and a target milestone for closure.
- Every acceptance claim links to an evidence path inside `qa/`. Prose is not evidence.

9 Premiere / After Effects handoff package

The package is the product's last mile. It is judged by whether a different person, on a different machine, opens it and can still edit.

9.1 Package contents

COMPONENT	REQUIREMENT
Transcript and timings	Accepted transcript with word timings; raw history included but clearly separated
Captions	SRT and/or ASS per ratio; if not applicable, an explicit `NOT_APPLICABLE` marker
Timeline artifacts	`scenes.json`, `cuts.json`, `master_edl.json` with source→output mapping
Visuals	`visuals/` folder, approved assets only, relative and portable paths
Asset metadata	`assets.json` with hash, source, licence, confidence and review state
Project file	Supported XML or project file, limited to the declared app version and effect subset
Reference material	Preview video, render profile, font list, dependencies, licences
Records	Manifest, QA report, logs, approvals, provenance, cost record

9.2 Import acceptance test

1. Open the sequence in the **actual** declared Premiere build — not a re-import of the render.
2. Confirm **zero missing media** and **zero missing fonts**.
3. Confirm timing matches the approved preview within one frame at three checkpoints.
4. Confirm each element declared editable is genuinely editable (trim, retime, recaption, replace).
5. Relocate the package (move, zip/extract, change drive letter) and repeat steps 1–2.
6. Record the result in `qa/import_log` with app version and build number.

> **TWO CLAIMS THAT MUST NEVER BE BLURRED**
>
> **Generic XML/EDL is not a native project.** It may import; it does not guarantee native effects, captions or motion.
>
> **Premiere and After Effects need separate adapters.** A passing Premiere test says nothing about AE, and vice versa.

10 Compute and environment strategy

Heavy processing runs on Kaggle; everything cheap and deterministic runs locally. GPU availability is treated as unproven until measured in the specific run that used it.

WORKLOAD	BASELINE	GUIDANCE	DEFAULT CLAIM
Probe, schemas	CPU, light	Run locally and on Kaggle first; these are the fast regression tests	`PASS`
ASR	CPU possible, GPU may help	Benchmark model and runtime separately; record the exact model size used	`NOT_TESTED`
Alignment / gap recovery	Tool and language dependent	Verify Hindi support on a small sample before trusting it on a full run	`NOT_TESTED`
Scene / OCR	Usually CPU	Fixture and representative-clip tests; no GPU assumed	`NOT_TESTED`

WORKLOAD	BASELINE	GUIDANCE	DEFAULT CLAIM
Face tracking	Backend and device dependent	Test MediaPipe on the target Windows machine and on Kaggle; if unproven, the check stays NOT_TESTED	NOT_TESTED
Motion render	CPU / RAM / Chromium dependent	Small templates only at first; measure speed, frame count and licence per template	NOT_TESTED
Image generation	Often heavy, model dependent	Optional; API disabled by default; no assumption that a consumer GPU fits the model	NOT_APPLICABLE
Premiere import	Local target application	Not possible on Kaggle; requires the local machine and a recorded build number	NOT_TESTED

10.1 Recorded per run

- Model and tool versions, dependency set, licences, device and backend used.
- Cache state; peak RAM and VRAM; timings for download, load, decode, inference, render and QA.
- Actual CUDA device selected, when a GPU is used at all.
- Source and output duration, config hash, exit code, log path, artifact hashes, QA status.
- Windows UTF-8 logging and non-ASCII path handling tested explicitly.
- Speed extrapolated only from natural 10–15 min sources, never from short clips.

> **COST DISCIPLINE**
>
> Free-tier compute is finite and variable. A run that depends on a paid or quota-limited resource must declare it at Gate 1, cap it, cache it and record actual cost in the manifest. Silent spend is a P0 defect.

11 Security, rights and recovery

11.1 Security

- Originals are read-only and content-hashed; each run occupies a unique isolated folder.
- Validate paths, IDs, schemas, time bounds, asset references and renderer capabilities before execution — not during.
- Treat all generated text as untrusted input; never execute shell commands derived from model output by default.
- Secrets never enter logs, manifests, bundles or source control; scan artifacts before packaging.
- Network retries are bounded to two backoff attempts; a deterministic failure stops the run instead of looping.

11.2 Rights and licensing

- Verify licences separately for footage, music, fonts, models, renderers and services — one clearance does not cover another.
- Paid APIs and models are off by default; enabling one requires explicit approval, a spend cap, caching and provenance.
- Every asset records source, licence and reviewer; unknown provenance is a blocking state, not a warning.
- AI-generated assets are labelled as illustrations in both the manifest and the UI.

11.3 Recovery

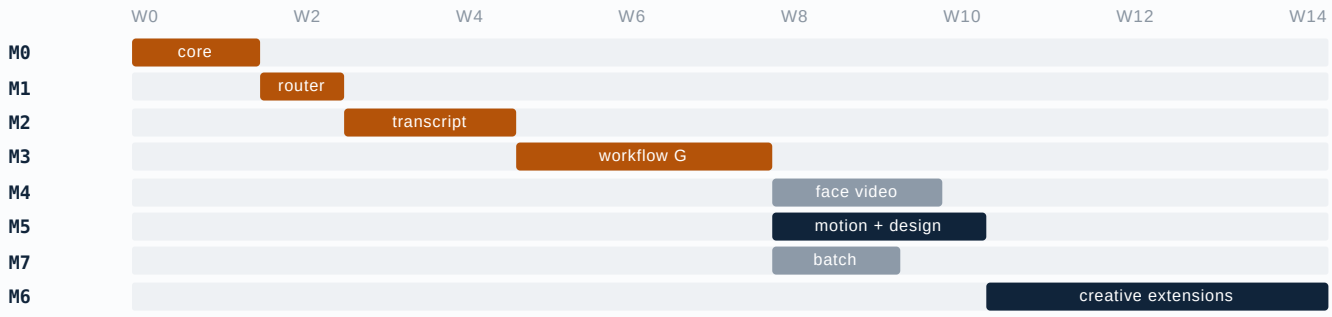
- Resume only from the last validated checkpoint. A partial output is invalid as cache and must be deleted or quarantined.
- Every fallback — safe crop, placeholder caption, approved substitute — is visible in UI, manifest and QA report.
- Cancellation is tested at each stage transition, not only at the start.
- No automatic publishing, no unapproved face swap, no voice cloning.

12 Build sequence, milestone plan and critical path

Modules are built independently against fixed fixtures and stable contracts, then integrated against real upstream outputs. Durations below are indicative effort for one primary builder working with review support; calendar dates are fixed at kickoff, not invented here.

MS.	SCOPE	EFFORT	DEPENDS ON	EXIT EVIDENCE
M0	Core contracts: IDs, schemas, manifests, logging, recovery, validation	1.5 wk	—	Clean tests; corruption and cancel tests; a run reproducible from its manifest alone
M1	Input router: audio, face, no-face, mixed, corrupt handling	1 wk	M0	Four fixture classes routed deterministically; override path and corrupt-file test
M2	Transcript intelligence: ASR draft, coverage, targeted retry, correction	2 wk	M0, M1	Reference sample with audit; raw history preserved; diff log complete
M3	**Workflow G**: scene plan, Gate 1, visuals, captions, preview, Premiere package	3 wk	M2	Full QA on a 10–15 min source; actual Premiere import with zero missing media; KPI baseline recorded
M4	Face video path: F04, F06, F01, F05	2 wk	M3	Three clips across ratios; subject-loss and jitter tests; fallback documented
M5	Motion and design baseline: F16, F20 template, F19-lite, F13, F15	2.5 wk	M3	Repeatable renders; font and licence QA; three design outputs
M6	Creative extensions: F07, F08, F09, F10, F11, F12, F14, F18, F21, F23	4 wk	M4, M5	Per-feature acceptance evidence; paid opt-in verified with a capped test call
M7	Batch and orchestration: F17 queue, status, workflow management	1.5 wk	M3	Failure isolation, cancellation, approval enforcement, regression suite

12.1 Critical path and sequencing rule



Dark amber = critical path (M0 → M1 → M2 → M3 → M5 → M6, ≈ 14 weeks). Grey = parallel work that may slip without blocking the path. Sequential effort if nothing overlaps ≈ 17.5 weeks.

- **Nothing starts before its dependency passes.** A milestone may be partially closed, but the next one begins only on the exit evidence of the previous.
- **Extensions never block the primary path.** If M6 slips, M3/M4/M5 remain shippable, because they do not import extension modules.
- **M3 is the decision point.** If Workflow G cannot reach a clean Premiere import within its budget, the correct response is to narrow scope — not to add features.

12.2 Definition of Ready (per milestone)

A milestone may start only when: its contracts are frozen and versioned; fixtures exist and are hash-recorded; the target hardware or app build is named; the reviewer and approver roles are assigned; and the exit evidence is written down *before* building.

12.3 Roles

ROLE	ACCOUNTABLE FOR	CANNOT SELF-APPROVE
Builder	Modules, tests, artifacts, logs, honest status reporting	Own milestone exit evidence
Reviewer / approver	Gate 1 and Gate 2 decisions; transcript acceptance; rights posture	Code they wrote in the same run
Product owner	Scope, milestone order, paid spend, KPI targets	QA results
End user (editor)	Final creative assembly in Premiere; publishing decisions	—

With a single-person team these roles still exist as separate hats, and the separation is recorded per approval in `approvals.json` — otherwise Gate 1 and Gate 2 collapse into a formality.

13 Risk register

The eight risks below decide whether this roadmap is realistic. Each has a trigger so it is detected early rather than discovered at a release gate.

#	RISK	L / I	EARLY TRIGGER	MITIGATION
R1	Hindi/Hinglish ASR coverage and spelling fall short of target on real voiceovers	H / H	Quick-sample accuracy below 90% before correction on the first natural source	Keep raw layer + human correction as the product promise; benchmark 2–3 engines; never claim "automatic accuracy"; make correction fast and diff-based
R2	Premiere import fidelity: XML imports but captions, effects or motion fall back to flat media	M / H	First import log shows substituted or dropped elements	Define the supported subset explicitly per app build; declare non-editable elements honestly; keep SRT/ASS as the guaranteed caption path
R3	Face tracking unstable on the target Windows machine; jitter or subject loss	M / M	Tracking test shows > 5% frames without subject	F06 is extension-tier; safe crop fallback with visible label; per-device test before any claim
R4	Compute and quota limits (Kaggle sessions, GPU variability, timeouts)	H / M	Runs failing on session timeout or memory ceiling	Checkpoint per stage; chunk long media; cache models; measure peak RAM/VRAM per run
R5	Font and asset licensing blocks commercial delivery	M / H	Brand font has no commercial licence on record	Licence field is mandatory in `brand.json` and `assets.json`; free-licence fallback font shipped; audit before M3
R6	Scope creep from extension features pulling effort away from Workflow G	H / H	Any M6 work started before M3 exit evidence	Tier system treats this as a process violation; the critical path is reviewed at every milestone gate
R7	Devanagari rendering and subtitle sync break at unusual sizes or mixed scripts	M / M	Any broken glyph in the audit, or concatenated words at line breaks	Test matras, conjuncts, digits and mixed-script pairs at three canvas sizes; render preset previews before editing
R8	Approval gates become rubber stamps under delivery pressure	M / H	Approvals recorded without a hash, in bulk, or by the same person who built the artifact	Hash-bound approvals, separate approver role, publication of approval turnaround in the run manifest

L / I = likelihood / impact: H high, M medium, L low. Re-score at each milestone exit.

14 Operational checklist and Definition of Done

The checklist is per run; the Definition of Done is per milestone. They are kept together so that "done" and "operable" cannot drift apart.

14.1 Per-run checklist

STAGE	CHECKLIST
Before run	Rights confirmed for every input; media and brief present; output ratio(s) chosen; target app build recorded; disk and memory checked
During run	Run ID allocated; versions and config hash recorded; checkpoints written per stage; paid actions confirmed at Gate 1 only; logs streaming
After run	QA executed; hashes and manifest verified; approvals archived; cost recorded; status published honestly; package archived with checkpoint

14.2 Definition of Done

- Clean install with pinned dependencies and documented commands.
- Clear logs and status for both valid and invalid inputs.
- Source files unchanged; run ID and manifest present for every run.
- Feature outputs pass schema and path validation, and survive relocation.
- All applicable QA checks pass, or the limitation is explicitly accepted and recorded with an owner.
- No hidden paid actions, and no silent substitutions anywhere in the pipeline.
- Recovery and cancellation tested; incomplete outputs rejected as cache.
- Validated on natural full-length media (10–15 min), not teasers or test tones.
- Actual app and device checks complete, or explicitly marked `NOT_TESTED`.
- A reviewer can reproduce the result from inputs, settings, versions and written instructions alone.

15 Canonical workflow chart

One chart, aligned to the stage IDs in §3.3. Gate markers show where a human decision is mandatory; nothing after a gate may proceed on assumption.

```
START
  |
S0  ├─ INTAKE: media + brief + output ratio
S1  ├─ VALIDATE + HASH + RUN_ID + ISOLATED WORKSPACE
S2  ├─ INPUT ROUTER: audio_only | video_face | video_no_face | mixed | unknown
S3  ├─ ANALYSIS: transcript + coverage + scene / audio / face analysis
S4  ├─ CORRECTION REVIEW ──────────────► ACCEPTED TRANSCRIPT (canonical)
S5  ├─ PLANNER: scenes + cuts + assets + captions + Master EDL / design spec
S6  ├─ ◆ GATE 1 — approve plan, cuts, assets, rights, paid actions
S7  ├─ EXECUTE: FFmpeg | motion | design | NLE adapter
S8  ├─ PREVIEW + QA: glyphs, sync, safe zones, loudness, timing, rights
S9  ├─ ◆ GATE 2 — approve preview and final settings
S10 ├─ FINAL RENDER / PACKAGE
S11 ├─ FULL QA (incl. actual Premiere import test)
S12 ├─ EXPORT + MANIFEST + LOGS + ARCHIVE / CHECKPOINT
  |
 END ── publishing is separate and never automatic
```

15.1 Router branches

BRANCH	BEHAVIOUR
Audio-only	Approved visuals and brand card; no face detection, no crop tracking
Face video	Preserve footage; optional subject-follow reframe for 9:16 and 1:1 with documented fallback
No-face video	Preserve footage with safe crop or pad; F06 off by default
Mixed / low confidence	Flag uncertain segments and request review; no automatic branch change after approval
Image / design	Skip transcript and EDL stages; mark them `NOT_APPLICABLE` rather than omitting them

16 Traceability matrix

Each workflow, its features, its milestone and its mandatory gate on one line. Use this to answer "what does M3 actually include?" without re-reading the whole document.

WF	WORKFLOW	FEATURES USED	MILESTONE	MANDATORY GATES	PRIMARY QA AREA
A	Talking head → Reel	F04, F06, F01, F02, F05	M4	G1, G2	Reframe, subtitles, audio
B	Screenshot → Motion	F19, F20	M5	G1, G2	Motion, assets
C	Podcast → Shorts	F07, F03, F10, F06, F01	M6	G1, G2 + action gate	Cuts, assets, subtitles
D	Brief → Design pack	F16, F13, F15, F14	M5	G1, G2	Assets, export
E	Lyrics → Kinetic text	F21, F20, F23	M6	G1, G2 + action gate	Motion, lyric alignment
F	Reference-inspired edit	F09, F03, F10, F01, F20	M6	G1, G2	Assets and rights
G	**Voiceover → Premiere**	F01, F02, F03, F04, F05, F12	**M3**	G1, G2	Transcript, subtitles, export, Premiere
H	Image operation	F14	M5	G2	Assets, export
I	Batch	F17 + any	M7	per-job G1, G2	Recovery, isolation

16.1 Using this matrix at a milestone gate

- **Scope check:** a milestone may not be signed off while a workflow it owns has an untested feature in its row.
- **Gate check:** if a workflow can reach export without passing its listed gates, the gate is decorative and must be fixed before the next milestone.
- **QA check:** the primary QA area names the only checks that can block that workflow — everything else is advisory for it.
- **Extension check:** any feature used by a single workflow only is by definition an extension and cannot enter the critical path.

17 Open decisions, glossary and change control

17.1 Open decisions to close before M2

#	DECISION	OPTIONS	WHY IT BLOCKS
D1	Which Premiere build(s) are declared supported	Single exact build vs a version band	F12 acceptance and every import claim depend on it
D2	Target Windows machine specification	Named CPU/GPU/RAM baseline	Face tracking and motion render claims are device-specific
D3	ASR engine and model size	Faster-Whisper sizes; hosted alternative	Coverage and accuracy targets cannot be set without it
D4	Captions burned-in vs delivered as tracks	Burn-in, sidecar, or both per workflow	Changes F01/F05 acceptance and the package contract
D5	Paid budget ceiling for F23 and F09	Zero, per-run cap, or monthly cap	Gate 1 cannot approve paid actions without a ceiling
D6	Who formally approves gates	Solo owner with recorded separation, or named reviewer	Gate semantics are meaningless without an accountable approver
D7	Brand and fallback font licences	Licensed brand font vs open-licence substitute	Blocks M3 packaging for commercial delivery

17.2 Glossary

TERM	MEANING IN THIS DOCUMENT
ASR	Automatic speech recognition — produces the draft transcript, never the canonical one
EDL / Master EDL	The declarative edit: source→output mapping, tracks, crop keyframes, asset references
Gate 1 / Gate 2	Human approvals at S6 (plan) and S9 (preview), each bound to an artifact hash
LUFS / dBTP	Integrated loudness / true peak — the audio acceptance measures
WER	Word error rate: (substitutions + deletions + insertions) ÷ reference words
Coverage gap	A region where speech is detected but no words exist; flagged above 2 s
NOT_TESTED	Capability exists in specification but has no execution evidence on the target device or app
NOT_APPLICABLE	Check does not apply to this workflow; the reason is recorded, not implied
Portable package	Output set that survives being moved, zipped and relinked without losing media, fonts or editability
Tier (Core / Extension / Reserved)	Build order and blocking status, not perceived importance

17.3 Change control

- **Schema changes** require a version bump plus a migration note; a breaking change requires re-running affected acceptance evidence.
- **New features** enter as Extension tier with a named workflow and milestone; no feature enters without a workflow to serve.
- **Target changes** (KPI numbers, thresholds, declared app builds) require a recorded decision with the reason and the date.
- **Re-scoring risks** happens at every milestone exit; an unchanged risk register after a milestone is treated as an unreviewed one.
- **Version history** is kept in the manifest of this document, not in filenames alone.

Product statement. Kang Automation is a modular, approval-driven creative workspace that converts user-provided media and briefs into reviewable plans, rendered assets and portable editing packages. Reliability is measured by reproducible artifacts and explicit QA evidence — not by this roadmap, a model name, or a process exit code alone.